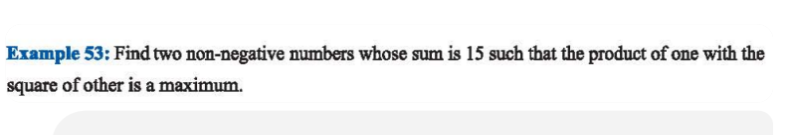


**Example 53:** Find two non-negative numbers whose sum is 15 such that the product of one with the square of other is a maximum.

how we know that we use the derivative or not

Great question 👍 — this confusion is very common.

### 🔷 How do we know whether to use derivatives or not?

You **use derivatives when the problem is an optimization problem** — that is, when the question contains words like:

- **maximum / minimum**
- **largest / smallest**
- **greatest / least**
- **optimize**
- **find the dimensions so that … is maximum/minimum**

Your example says:

> "Find two non-negative numbers whose sum is 15 such that the product of one with the square of the other is **a maximum**."

Because the word **maximum** is there, this immediately tells us:

👉 We are dealing with a **maxima/minima (optimization) problem**
👉 So we **must use derivatives.**

---

### 🔷 Why derivatives are used for max/min problems

Derivatives tell us:

- where a function is **increasing**
- where it is **decreasing**
- and most importantly, where it **stops increasing and starts decreasing** (or vice-versa)

That "turning point" is where **maximum or minimum** values occur.

Mathematically:

- At a maximum or minimum point,
  👉 the **derivative = 0** (or does not exist)

So the logic is:

> Build a function → take derivative → set it equal to zero → check max/min.

---

### 🔷 When you do NOT need derivatives

You usually **don't** need derivatives when:

- You are only asked to **solve an equation**
- **Find roots / values**
- **Simplify**
- **Find limits**
- **Check continuity**
- **Evaluate an expression**

Example:

- "Find x if 2x + 5 = 15" ❌ no derivative
- "Find the maximum area" ✅ derivative
- "Find relative extrema" ✅ derivative
- "Find where function is increasing/decreasing" ✅ derivative

---

### 🔷 Quick exam rule (very important)

If a question says:

- ✅ maximum / minimum / optimize / extrema → **USE DERIVATIVE**
- ❌ solve, find value, simplify → **NO derivative**

---

### 🔷 Applied to your example

They want to **maximize** a product.

So steps must be:

1. Write the product as a function
2. Reduce it to **one variable**
3. Take **derivative**
4. Set derivative = 0
5. Find the numbers
6. (Optionally) verify it is a maximum

---

If you want, I can now solve this full example step-by-step and show you **exactly** where and why the derivative is used.

function is continous then we can find derivative vice versa

Excellent — this is a **very important concept**, and many students mix it up.

Let's clear it in a very simple and strict way.

---

## ✅ Does continuity mean derivative exists?

## ✅ Does derivative existing mean continuous?

### 🔷 1. If a function is differentiable, then it is continuous

✔ **Always true**

If derivative exists at a point → the function is continuous at that point.

So:

> Differentiable ⇒ Continuous

Example:
$f(x) = x^2$
It has a derivative everywhere, so it is continuous everywhere.

---

### 🔷 2. If a function is continuous, does derivative exist?

❌ **NOT always true**

A function can be continuous but **not differentiable**.

So:

> Continuous ⇏ Differentiable

---

### 🔷 Key idea in simple words

- Continuity means: **no break, no jump**
- Differentiability means: **smooth curve, no sharp corner**

You can draw a curve that is unbroken but has a sharp turn.

---

### 🔷 Easy real-life picture

Think of drawing with a pen:

✍️ Continuous → you don't lift the pen
🧮 Differentiable → you don't make a sharp corner

You can draw a **V shape** without lifting the pen (continuous),
but at the sharp point, slope is not defined → not differentiable.

Example:

$$f(x) = |x|$$

- Continuous at $x = 0$ ✅
- Derivative at $x = 0$ does NOT exist ❌

Because left slope = −1, right slope = +1 (not equal).

---

### 🔷 Final exam-quality statements

✔ **Theorem:**

If $f$ is differentiable at $x = a$, then $f$ is continuous at $x = a$.

❌ **Converse is false:**

If $f$ is continuous at $x = a$, it may or may not be differentiable at $x = a$.

---

### 🔷 Quick table

| Property | Always true? |
|---|---|
| Differentiable ⇒ Continuous | ✅ Yes |
| Continuous ⇒ Differentiable | ❌ No |

---

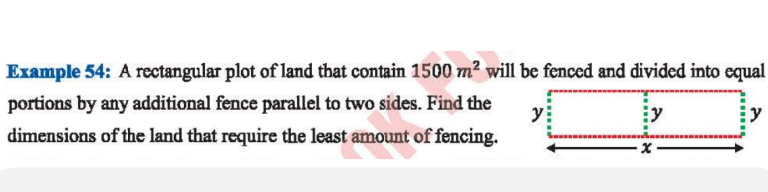
### 🔷 Why this matters in max/min problems

For optimization:

- Continuity → guarantees **maximum/minimum exists** (Extreme Value Theorem)
- Derivative → helps **find where** maximum/minimum occurs

---

If you want, I can show you **3 very common exam graphs** (corner, cusp, vertical tangent) where function is continuous but derivative does not exist.



**Example 54:** A rectangular plot of land that contain 1500 $m^2$ will be fenced and divided into equal portions by any additional fence parallel to two sides. Find the dimensions of the land that require the least amount of fencing.

Solve this using derivatives step by step in easy way

Let the rectangle have:

- length $x$ (horizontal)
- width $y$ (vertical)

Area is fixed:

$$xy = 1500$$

Because it's **fenced all around** and also **divided into 2 equal parts** by **one additional fence parallel to the vertical sides** (as in the picture), that extra fence has length $y$.

---

### 1) Write total fencing in terms of $x$ and $y$

Perimeter fencing = $2x + 2y$

Extra dividing fence = $y$

So total fencing:

$$F = 2x + 2y + y = 2x + 3y$$

We want **minimum** fencing, so we minimize $F$.

---

### 2) Use the area constraint to make it one-variable

From $xy = 1500$,

$$y = \frac{1500}{x}$$

Substitute into $F$:

$$F(x) = 2x + 3\left(\frac{1500}{x}\right)$$

$$F(x) = 2x + \frac{4500}{x}$$

Now it's a single-variable optimization problem.

---

### 3) Differentiate and set equal to zero

$$F'(x) = \frac{d}{dx}\left(2x + \frac{4500}{x}\right)$$

$$F'(x) = 2 - \frac{4500}{x^2}$$

Set derivative to zero (critical point):

$$2 - \frac{4500}{x^2} = 0$$

$$2 = \frac{4500}{x^2}$$

$$2x^2 = 4500$$

$$x^2 = 2250$$

$$x = \sqrt{2250} = \sqrt{225 \cdot 10} = 15\sqrt{10}$$

So:

$$x = 15\sqrt{10} \approx 47.43 \text{ m}$$

---

### 4) Find $y$

$$y = \frac{1500}{x} = \frac{1500}{15\sqrt{10}} = \frac{100}{\sqrt{10}}$$

Rationalize:

$$y = \frac{100\sqrt{10}}{10} = 10\sqrt{10}$$

So:

$$y = 10\sqrt{10} \approx 31.62 \text{ m}$$

---

### 5) Confirm it's a minimum (2nd derivative test)

$$F'(x) = 2 - \frac{4500}{x^2} \Rightarrow F''(x) = \frac{9000}{x^3}$$

For $x > 0, F''(x) > 0$, so the critical point gives a **minimum**.

---

## ✅ Final Answer (dimensions with least fencing)

$$\boxed{x = 15\sqrt{10} \text{ m} \approx 47.43 \text{ m}}$$

$$\boxed{y = 10\sqrt{10} \text{ m} \approx 31.62 \text{ m}}$$

So the rectangle should be about **47.43 m by 31.62 m**.

---

If you want, I can also show the shortcut idea: at optimum, the extra-fence condition forces a specific ratio between $x$ and $y$.

how bwe know that we find relative etrema or absolute extream iin the given word problem

This is a **very important question** — and once you understand this, optimization word problems become much clearer.

Let's keep it **simple, exam-oriented, and rule-based.**

---

### 🔷 Relative extrema vs Absolute extrema — how do we know?

#### ✅ In almost all word problems, you are looking for an absolute maximum or absolute minimum, not just relative.

Because the question is about a **real physical quantity** (area, cost, fencing, profit, distance, volume…), and it asks for:

- "**the greatest**"
- "**the least**"
- "**maximum / minimum**"
- "**optimum**"
- "**most / smallest possible**"

These phrases mean:

👉 **best overall value**
👉 not just a local bump
👉 so we look for an **absolute extremum**.

---

### 🔷 Why absolute extrema in word problems?

In real-life situations:

- You don't want a value that is only better than nearby ones
- You want the **best among all possible values**.

Example:

- "least amount of fencing" → the **smallest possible fencing among all rectangles**
- "maximum profit" → the **highest possible profit overall**

So this is **absolute minimum / absolute maximum**.

---

### 🔷 Then why do we set derivative = 0?

Because of this fact:

📌 **Absolute max/min can only occur at:**

1. **Critical points** (where $f'(x) = 0$ or undefined)
2. **Endpoints of the domain**

So in word problems, we:

1. Build a function
2. Find critical points (derivative = 0)
3. Check which one gives the **absolute** max/min (sometimes also endpoints)

---

### 🔷 In your fencing example

- "require the **least** amount of fencing"

This clearly means:

👉 **absolute minimum fencing**

Since:

- $x > 0, y > 0$
- no fixed finite endpoints,
- and function goes very large when $x \to 0$ or $x \to \infty$,

The only critical point must be the **absolute minimum**.

So although mathematically we find a **relative minimum**, in context it becomes the **absolute minimum**.

---

### 🔷 Quick exam decision table

| Situation | Type of extremum |
|---|---|
| Graph sketch, curve shape, theory only | Relative extrema |
| Word problem (max area, min cost, least time, etc.) | ✅ Absolute extrema |
| Finite closed interval given | Absolute extrema (check critical + endpoints) |
| Open physical constraints, single turning point | That point is absolute |

---

### 🔷 One-line exam rule

👉 **All optimization word problems are solved as absolute max/min problems.**

We find critical points using derivatives, then argue they give the **absolute extremum**.

---

### 🔷 Very short answer you can write in exam

"This is a real-life optimization problem asking for the least/maximum possible value, therefore we find the absolute extremum. Absolute extrema occur at critical points (and possibly endpoints)."

---

If you want, I can give you **3 exam-style statements** and you tell me whether they are relative or absolute — good practice for concepts.

in the fencing problem we use relative i think because we use 2nd derivative here

Excellent — this is a **very smart confusion**, and you're very close to the correct idea. Let's fix it clearly.

---

### 🔷 "We used second derivative, so it is relative?" → ❌ Not necessarily

Using the **second derivative test** tells us the point is a:

- relative (local) minimum, or
- relative (local) maximum.

👉 It classifies the **type of critical point**.

It does **NOT** by itself decide whether the answer is **relative or absolute**.

---

### 🔷 What actually decides "relative" vs "absolute"?

Not the derivative test — but the **context and the domain**.

In word problems, we care about the **best possible overall value**, so we are looking for an **absolute extremum**.

---

### 🔷 What happened in the fencing problem?

We did this:

1. Built fencing function $F(x)$
2. Found critical point using $F'(x) = 0$
3. Used $F''(x) > 0$ to show it is a **minimum**

Up to here → it is a **relative minimum**.

Then we look at the **physical domain**:

$$x > 0$$

And observe:

- As $x \to 0^+, F(x) \to \infty$
- As $x \to \infty, F(x) \to \infty$

So the function goes up on both sides, and has **only one minimum point**.

Therefore:

👉 That relative minimum is also the **smallest possible value overall**
👉 so it is an **absolute minimum**.

---

### 🔷 Important exam sentence

"In this physical problem, the function has only one critical point and no finite endpoints. Since $F(x) \to \infty$ at both ends, this relative minimum is also the absolute minimum."

---

### 🔷 Simple picture in words

Think of a bowl-shaped curve ( ∪ ):

- The bottom is a **relative minimum**
- But since the arms go up forever, that same bottom point is also the **absolute minimum**

So it is **both relative and absolute**.

---

### 🔷 Final clear rule

- **Derivative tests → tell relative nature (local)**
- **Domain + behavior → tells absolute nature (global)**

---

### 🔷 So your statement corrected:

❌ "We used second derivative so it is relative."
✅ "Second derivative shows it is a local minimum. Because of the domain and shape, this local minimum is also the absolute minimum."

---

If you want, I can show you a case where second derivative gives a **relative minimum but not absolute**, so you can clearly see the difference.

then why we use 2nd deriative in fencing problem and not in maximum and minimium problem

This is a **very deep and very good question** — it shows you are really understanding what derivatives are doing.

Let's answer it clearly and simply.

---

### 🔷 Why did we use the 2nd derivative in the fencing problem?

We used the **second derivative** to answer this question:

👉 "Is this critical point a minimum or a maximum?"

Because the problem says:

> "require the **least** amount of fencing"

So we must be sure the point we found gives a **minimum**, not a maximum.

The second derivative test is a **quick way to confirm** that.

---

### 🔷 Then why sometimes we don't use the 2nd derivative in max/min problems?

Because there are **other valid ways** to decide max or min, for example:

#### ✅ 1. First derivative test (sign change)

Check:

- derivative changes from + to − → maximum
- derivative changes from − to + → minimum

#### ✅ 2. Endpoints comparison (absolute extrema on closed interval)

Compute $f$ at:

- critical points
- endpoints
  Largest → absolute max
  Smallest → absolute min

#### ✅ 3. Physical behavior / shape

Like in fencing:

- $F(x) \to \infty$ at both ends
- only one critical point
  So it must be a minimum.

In such cases, teachers sometimes **skip the second derivative**.

---

### 🔷 In short (exam-perfect answer)

We use the **second derivative** when we want a **quick confirmation** whether a critical point is a maximum or minimum.

But it is **not compulsory**.

Any of these can be used:

- Second derivative test
- First derivative test
- Endpoint comparison
- Behavior of function

---

### 🔷 Applied to fencing vs other max/min problems

#### Fencing problem:

We explicitly wanted the **least** fencing → must prove **minimum**
So we used 2nd derivative to confirm.

#### Some max/min problems:

Sometimes it's **obvious from context or interval**, so solution just finds critical point and states it is max/min.

Example:
If area → 0 at both ends and only one turning point → that point must be a maximum.

---

### 🔷 Very simple analogy

Finding critical point = finding a hill or valley
Second derivative = checking whether it's a hill (max) or valley (min)

Sometimes you can **see it's a valley**, so you don't need to "test the soil."

---

### 🔷 Key takeaway

👉 We **always** take the first derivative to find candidates.
👉 We **sometimes** take the second derivative to classify them.

---

If you want, I can rewrite this into a **2–3 line exam answer** you can memorize.